

```
In [1]: #Numpy Crash course
```

```
In [2]: print("Numpy Crash course")
```

Numpy Crash course

```
In [4]: import numpy as np
```

```
In [6]: np.__version__
```

```
Out[6]: '1.26.4'
```

```
In [7]: install numpy
```

```
Cell In[7], line 1
      install numpy
      ^
SyntaxError: invalid syntax
```

```
In [8]: pip install numpy
```

Requirement already satisfied: numpy in c:\users\jagatjyoti\anaconda3\lib\site-packages (1.26.4)
Note: you may need to restart the kernel to use updated packages.

```
In [9]: np.amin
```

```
Out[9]: <function amin at 0x00000259413B8E30>
```

```
In [11]: from numpy import * # we are calling all 256 functions together at once
```

```
In [12]: my_list =[0,1,2,3,4,5]
```

```
Out[12]: [0, 1, 2, 3, 4, 5]
```

```
In [13]: type(my_list)
```

```
Out[13]: list
```

```
In [14]: arr =np.array(my_list)
```

```
In [15]: arr
```

```
Out[15]: array([0, 1, 2, 3, 4, 5])
```

```
In [19]: type(arr)
```

```
Out[19]: numpy.ndarray
```

```
In [23]: np.arange(10)
```

```
Out[23]: array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])
```

```
np.arange(10,20)
```

```
In [24]: np.arange(10,20,3)
```

```
Out[24]: array([10, 13, 16, 19])
```

```
In [25]: np.arange(5)
```

```
Out[25]: array([0, 1, 2, 3, 4])
```

```
In [26]: np.arange(10,20)
```

```
Out[26]: array([10, 11, 12, 13, 14, 15, 16, 17, 18, 19])
```

```
In [27]: np.zeros(5)
```

```
Out[27]: array([0., 0., 0., 0., 0.])
```

```
In [29]: np.zeros(5,dtype=int) # 1D Arrays
```

```
Out[29]: array([0, 0, 0, 0, 0])
```

```
In [31]: np.zeros((2,2)) # 2D Arrays
```

```
Out[31]: array([[0., 0.],  
               [0., 0.]])
```

```
In [32]: np.zeros((2,2), dtype=int) # 2D Arrays 2x2 -Matrix
```

```
Out[32]: array([[0, 0],  
               [0, 0]])
```

```
In [33]: np.zeros((5,10), dtype=int) # 2D Arrays 5x10 -Matrix -ND - Array
```

```
Out[33]: array([[0, 0, 0, 0, 0, 0, 0, 0, 0, 0],  
               [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],  
               [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],  
               [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],  
               [0, 0, 0, 0, 0, 0, 0, 0, 0, 0]])
```

```
In [35]: np.ones(10)
```

```
Out[35]: array([1., 1., 1., 1., 1., 1., 1., 1., 1., 1.])
```

```
In [36]: np.ones(10, dtype=int)
```

```
Out[36]: array([1, 1, 1, 1, 1, 1, 1, 1, 1, 1])
```

```
In [38]: np.ones((3,3), dtype=int)
```

```
Out[38]: array([[1, 1, 1],  
               [1, 1, 1],  
               [1, 1, 1]])
```

```
In [39]: np.twos((3,3), dtype=int)
```

```

-----
AttributeError                                Traceback (most recent call last)
Cell In[39], line 1
----> 1 np.twos((3,3), dtype=int)

File ~\anaconda3\Lib\site-packages\numpy\__init__.py:333, in __getattr__(attr)
    330     "Removed in NumPy 1.25.0"
    331     raise RuntimeError("Tester was removed in NumPy 1.25.")
--> 333 raise AttributeError("module {!r} has no attribute "
    334                        "{!r}".format(__name__, attr))

AttributeError: module 'numpy' has no attribute 'twos'

```

```
In [42]: np.random.rand(4) #Number suffel on each time we execute
```

```
Out[42]: array([0.47175612, 0.36359559, 0.58286456, 0.96163164])
```

```
In [43]: np.random.rand(4,4)
```

```
Out[43]: array([[0.85935364, 0.69864086, 0.54488157, 0.89383591],
                [0.23566029, 0.22094975, 0.75588557, 0.92956284],
                [0.07856675, 0.95875494, 0.92815902, 0.14554394],
                [0.88856586, 0.85144448, 0.94181649, 0.97086654]])
```

```
In [46]: np.random.randint(3) # it will print values from 0 to 3-1 that is between 0-2
```

```
Out[46]: 0
```

```
In [51]: np.random.randint(1,32) # it will print values between 0- 31
```

```
Out[51]: 6
```

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In [ ]:
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In [ ]:
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In [ ]:
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